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Resting-state fMRI Analysis of Alzheimer's Disease Progress Using Sparse Dictionary Learning

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***Abstract*—A novel data-driven resting state fMRI analysis based on sparse dictionary learning is presented. Although ICA has been a popular data-driven method for resting state fMRI data, the assumption that sources are independent often leads to a paradox in analyzing closely interconnected brain networks. Rather than using independency, the proposed approach starts from an assumption that a temporal dynamics at each voxel position is a sparse combination of global brain dynamics and then proposes a novel sparse dictionary learning method for analyzing the resting state fMRI analysis. Moreover, using a mixed model, we provide a statistically rigorous group analysis. Using extensive data set obtained from normal, Mild Cognitive Impairment (MCI), Clinical Dementia Rating scale (CDR) 0.5, CDR 1.0, and CDR 2.0 patients groups, we demonstrated that the changes of default mode network extracted by the proposed method is more closely correlated with the progress of Alzheimer disease.**

***Index Terms*—Data-driven functional magnetic resonance imaging (fMRI) analysis, K-SVD, sparse dictionary learning, statistical parametric mapping, resting-state, Alzheimer's disease.**

I. Introduction

Many researches in early days of neuroimaging field relied on a generalized linear model (GLM) [1]–[3], which were defined by certain hypotheses of the experiment. However, as the biological properties over brain regions and between a subject may vary, the use of canonical hemodynamic response functions in GLM can cause bias in statistical inference. To address this issue, many researchers have investigated data driven analysis methods that focus on exploring the data to extract data specific functional dynamics. Such data driven analysis methods include principal component analysis (PCA) and independent component analysis (ICA) [4] as representative examples. In particular, these data driven methods are well suited for studying brain at rest [5], since there is no pre-defined paradigm for resting state brain.

It is now well-known that these methods can extract default mode network (DMN) from resting state brain. DMN includes the brain regions of medial prefrontal cortex (MPFC), posterior cingulate cortex (PCC), left lateral parietal (LLP), and right lateral parietal (RLP). In humans, the default network has been considered to generate spontaneous thoughts at rest and it has been hypothesized that weakening DMN may be related to disorders including Alzheimer's disease, autism, and schizophrenia [6].

However, as the concept of brain functional connectivity can only be reasonable based upon certain level of dependence between signals in the brain, algorithms such as ICA based on the independence of signals has fundamental limiations in analyzing brain connectivity. More interestingly, it has been demonstrated that the success of ICA for resting state data analysis is due to their ability to handle sparse components rather than independent components [7]. Inspired by these finding, our group has developed a data driven fMRI analysis totally based on sparsity, and have demonstrated excellent activation detection in individual analysis results from a paradigm driven fMRI experiments [8]. In this paper, we further extend the results and provide a statistically rigorous mixed model group data analysis that is ideally suitable for resting-state fMRI analysis.

To confirm the validity of the proposed method, we provide extensive comparison using group data from normal, MCI, CDR 0.5, CDR 1.0 and CDR 2.0 scale Alzheimer subjects. Results indicate that extracted DMNs using the proposed method exhibit excellent correlation with the disease progress, whereas other existing methods such as seed-based, or ICA approach do not exhibit consistent changes of DMN patterns. Considering clinical findings of decreased metabolism in DMN with disease progress [9], we believe that the results in this paper provides strong indications that the proposed method is a powerful tool for resting state fMRI analysis.

II. Method

A. Data Acquisition

Resting state fMRI scans were obtained using a 3.0 T scanner (Model: Philips Intera Achieva, Phillips Healthcare, Netherlands). Scans involved the acquisition of 35 axial slices using a gradient echo planar imaging pulse sequence: TR= 3000 ms; TE= 35 ms; FOV (RL, AP, FH) = 220 mm x 140 mm x 220 mm; voxel size (RL, AP) = 2.875 mm x 2.875 mm. During scan, participants were instructed to lay still with eyes closed. A total of 100 acquisitions are obtained for each subject. We collected five groups of resting-state fMRI data: 1) normal of 22 subjects, 2) MCI of 37 subjects, 3) CDR 0.5 of 20 subjects, 4) CDR 1.0 of 27 subjects, and 5) CDR 2.0 of 13 subjects.

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B. Preprocessing

The images were first spatially realigned to remove movement artefact fMRI time-series. The images were then spatially normalized to a standard space, Montreal Neurological Institute (MNI) space, which is widely used by researchers and resampled with voxel size 2 mm x 2 mm x 2 mm. Spatial smoothing was then applied with full-width at half-maximum (FWHM) Gaussian kernel size 8 mm x 8 mm x 8 mm. The brain region of functional data was extracted using a segmented anatomy data as a mask image with respect to gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF). We used a discrete cosine transform (DCT) filter with a cutoff frequency of $1/128$ Hz, which is an appropriate range of frequency for resting-state data, which show low frequency oscillations with the range of $0.0 - 0.1$ Hz, in general.

C. Data Analysis using Conventional Methods

We used two conventional methods for resting state fMRI analysis. First, Multi-session temporal concatenation of Multivariate Exploratory Linear Optimized Decomposition into Independent Components (MELODIC) within FMRIB's Software Library (FSL) [10] is used as an ICA methods. Second, we use Functional connectivity toolbox (conn) based on Statistical Parametric Mapping (SPM) for seed based analysis.

D. Mixed Model Sparse GLM

Data-driven sparse GLM with sparsity level k can be represented as [8]:

$$\mathbf{y}_i \quad = \quad \mathbf{D}_{I_i}\mathbf{x}_{I_i} + \boldsymbol{\varepsilon}_i, \; i = 1, \ldots, N \; , \tag{1}$$

where $\boldsymbol{y}_i \in \mathbb{R}^m$, $\mathbf{D}_{I_i} \in \mathbb{R}^{m \times k}$, $\mathbf{x}_{I_i} \in \mathbb{R}^k$, and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^m$ represent samples of a BOLD signal, the regressors, the corresponding response signal strength and the corresponding noise at the i-th voxel, respectively. Note that the model requires an estimation of a global dictionary $\mathbf{D}$, from which sparse subset index I_i needs to be estimated at each voxel. A rationale underlying this representation is that a temporal dynamics at each voxel can be represented by a sparse combination of global dynamics. We believe that this model can represent the nature of brain connectivity more closely, since a brain network is known to have small worldness property, so a temporal dynamics at each spatial location can be usually represented mostly with a temporal dynamics within the community in addition to a few long-range connections. To estimate the global dictionary D and membership indices I_i, the K-SVD algorithm was used for dictionary learning and a simple thresholding method was for a sparse coding method in an iterative manner [8], [11]. We used the following parameters for sparse dictionary learning process: 20 for the number of dictionary regressors, 3 for the sparsity level, and 5 for the number of iterations.

We are interested in finding group level statistics how much effect a specific regressor from global dictionary $\mathbf{D}$ has for each voxel of data. For this, we employ a mixed model that provides a unified framework for various fixed and random elements [12]. More specifically, individual data within each group was manipulated to have zero mean and unit variance, and then the data was temporally concatenated to built a group data $\mathbf{y}$. Then, a mixed model is written as

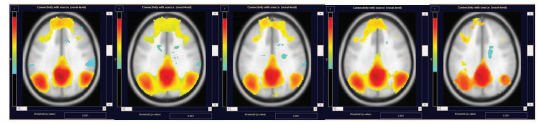

Fig. 1. Seed based analysis results using 'conn v.12' with PCC as a seed with p-value < 0.001. Each figure represents the group, normal, MCI, CDR 0.5, CDR 1.0, and CDR 2.0 from left to right.

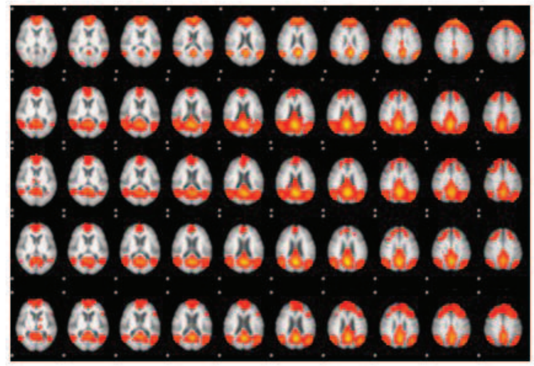

Fig. 2. ICA results using 'Melodic 3.0 of FSL' with a multi-session temporal concatenation and threshold level of 0.5. Each row of the images represents the group, normal, MCI, CDR 0.5, CDR 1.0, and CDR 2.0 from top to bottom. For each column, difference slices images are illustrated

$$\mathbf{y} \quad = \quad \mathbf{D}\mathbf{x} + \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} \tag{2}$$

where $\mathbf{y}$ is the group observations, $\mathbf{x}$ denotes a group level fixed effects, and $\boldsymbol{\gamma}$ is a random effect parameter following $N(\mathbf{0}, \mathbf{G})$, and $\boldsymbol{\varepsilon}$ is an additive noises that follows $N(\mathbf{0}, \mathbf{R})$. The fixed effect group design matrix $\mathbf{D}$ is constructed by concatenating individual design matrices, whereas the random effect design matrix $\mathbf{Z}$ can be constructed into a block diagonal structure obtained from individual design matrices. We use the maximum likelihood (ML) to obtain estimates of $\mathbf{G}$ and $\mathbf{R}$, and estimates of $\mathbf{x}$ and γ by solving the mixed model equations.

Finally, statistical inferences are obtained by testing the hypothesis

$$H : \mathbf{L} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\gamma} \end{bmatrix} \quad = \quad 0 \; . \tag{3}$$

If $\mathbf{L}$ is a matrix, a F-statistic can be established as

$$F \quad = \quad \frac{\begin{bmatrix} \hat{\mathbf{x}} \\ \hat{\boldsymbol{\gamma}} \end{bmatrix}' \mathbf{L}'(\mathbf{L}\hat{\mathbf{C}}\mathbf{L}')^{-1}\mathbf{L} \begin{bmatrix} \hat{\mathbf{x}} \\ \hat{\boldsymbol{\gamma}} \end{bmatrix}}{r} \tag{4}$$

where $r = rank(\mathbf{L}\hat{\mathbf{C}}\mathbf{L}')$ and

$$\hat{\mathbf{C}} \quad = \quad \begin{bmatrix} \mathbf{D}'\hat{\mathbf{R}}^{-1}\mathbf{D} & \mathbf{D}'\hat{\mathbf{R}}^{-1}\mathbf{Z} \\ \mathbf{Z}'\hat{\mathbf{R}}^{-1}\mathbf{D} & \mathbf{Z}'\hat{\mathbf{R}}^{-1}\mathbf{Z} + \hat{\mathbf{G}}^{-1} \end{bmatrix}^{-} \tag{5}$$

We used a random field corrected p-value as suggested in [13].

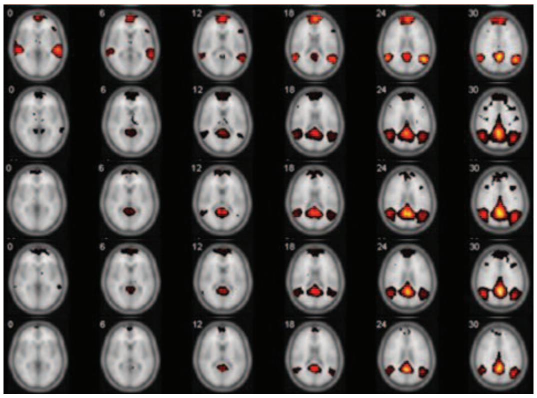

Fig. 3. The results using proposed method with p-value < 0.001. Each row of the images represents the group, normal, MCI, CDR 0.5, CDR 1.0, and CDR 2.0 from top to bottom. For each column, difference slices images are illustrated.

III. Results

Comparative analyses using seed based analysis, ICA, and the proposed method, were conducted using normal, MCI, CDR 0.5, CDR 1.0, and CDR 2.0. As in Fig. 1, the seed based analysis results shows the DMN patterns across all groups; however it does not reveal any noticeable differences between groups. Although the result of normal group indicates clear DMN activation pattern in both ICA (Fig. 2) and the proposed method (Fig. 3), in abnormal groups using ICA method decreasing tendency of DMN patterns from MCI to CDR 2.0 were not clearly revealed, whereas the proposed method provides results closely correlated with the course of AD, as in Fig. 3.

IV. Conclusion

In this article, we developed a mixed-model sparse dictionary learning framework for resting state fMRI analysis. Unlike the ICA methods, the new algorithm exploits that a temporal dynamics at each voxel can be represented as a sparse combination of global dynamics thanks to the property of small-worldness of brain network. Using mixed model, we also implemented statistically rigorous group analysis and inference tools. We compared our tools with the existing seed-based approach and ICA approach for normal, MCI and Alzheimer with different disease scale. The results indicated that DMN network extracted using our method is closely correlated with the progress of disease, indicating that the tool has great potential for resting state analysis.

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